

DATA CLEANING & PREPARATION REPORT

1. Objective

The objective of this project is to clean and prepare a raw e-commerce dataset by identifying missing values, removing duplicate records, correcting data formats, and preparing the dataset for further analysis using Python in Google Colab.

2. Tools and Technologies Used

- Python
- Pandas Library
- NumPy Library
- Google Colab

3. Dataset Description

The dataset used in this project contains e-commerce order details including Order ID, Date, Customer ID, Product, Quantity, Unit Price, Shipping Address, Payment Method, Order Status, Tracking Number, Items in Cart, Coupon Code, Referral Source, and Total Price. The dataset consisted of 1200 records and 14 columns.

4. Methodology / Steps Performed

1. Imported required libraries such as Pandas and NumPy.
2. Loaded the dataset into Google Colab.
3. Viewed the dataset using `df.head()`.
4. Checked dataset information using `df.info()`.
5. Identified missing values using `df.isnull().sum()`.
6. Replaced missing CouponCode values with 'No Coupon'.
7. Checked duplicate records using `df.duplicated().sum()`.
8. Verified duplicate Order IDs.
9. Corrected date formats using pandas.
10. Performed final verification to ensure clean data.

5. Sample Code Used

```
import pandas as pd
import numpy as np

df = pd.read_excel("Dataset for Data Analytics.xlsx")

df.isnull().sum()

df["CouponCode"].fillna("No Coupon", inplace=True)

df.duplicated().sum()
```

```
df["Date"] = pd.to_datetime(df["Date"])
```

```
df.to_csv("cleaned_dataset.csv", index=False)
```

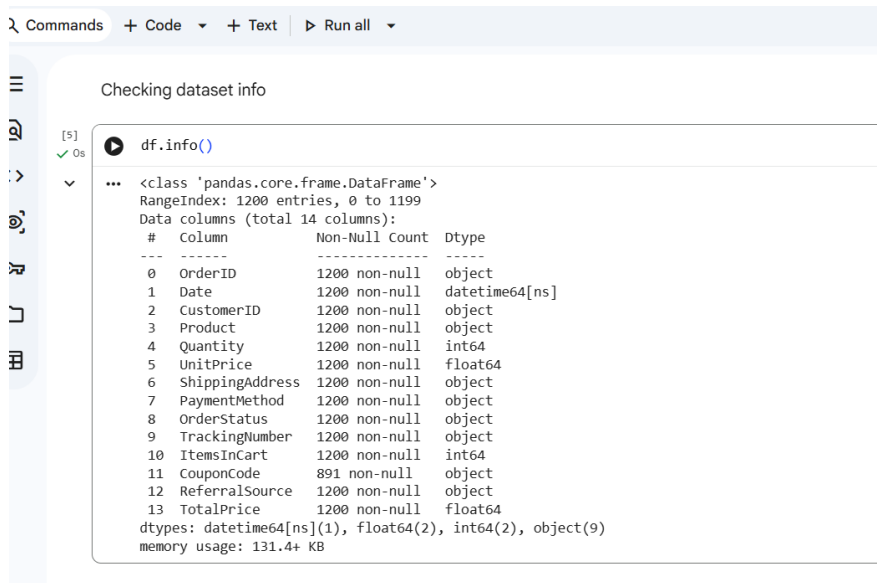
6. Results

- Successfully identified and handled missing values.
- Verified that no duplicate rows existed.
- Ensured unique Order IDs.
- Corrected date formats.
- Prepared a fully cleaned dataset for future analysis.
- Final verification confirmed zero missing values and zero duplicate records.

7. Conclusion

This project helped in understanding the importance of data cleaning and preparation in data analytics. Clean and structured data is essential for accurate analysis, reporting, and decision-making. The final cleaned dataset is ready for further analysis and visualization tasks.

8. Screenshots



The screenshot shows a Jupyter Notebook interface with a command bar at the top containing 'Commands', '+ Code', '+ Text', and 'Run all'. The notebook is titled 'Checking dataset info'. The code cell contains the command `df.info()`. The output shows the DataFrame's structure: 1200 entries, 14 columns. A table lists the columns with their non-null counts and dtypes. At the bottom, it shows the dtypes for each column and the total memory usage.

```
[5]: df.info()
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1200 entries, 0 to 1199
Data columns (total 14 columns):
#   Column                Non-Null Count  Dtype
---  -
0   OrderID               1200 non-null  object
1   Date                  1200 non-null  datetime64[ns]
2   CustomerID            1200 non-null  object
3   Product               1200 non-null  object
4   Quantity              1200 non-null  int64
5   UnitPrice             1200 non-null  float64
6   ShippingAddress       1200 non-null  object
7   PaymentMethod         1200 non-null  object
8   OrderStatus           1200 non-null  object
9   TrackingNumber        1200 non-null  object
10  ItemsInCart           1200 non-null  int64
11  CouponCode            891 non-null   object
12  ReferralSource        1200 non-null  object
13  TotalPrice            1200 non-null  float64
dtypes: datetime64[ns](1), float64(2), int64(2), object(9)
memory usage: 131.4+ KB
```

Before Cleaning the data:

```
df.isnull().sum()
```

...	0
OrderID	0
Date	0
CustomerID	0
Product	0
Quantity	0
UnitPrice	0
ShippingAddress	0
PaymentMethod	0
OrderStatus	0
TrackingNumber	0
ItemsInCart	0
CouponCode	309
ReferralSource	0
TotalPrice	0

dtype: int64

After Cleaning the data:

```
df.isnull().sum()
```

	0
OrderID	0
Date	0
CustomerID	0
Product	0
Quantity	0
UnitPrice	0
ShippingAddress	0
PaymentMethod	0
OrderStatus	0
TrackingNumber	0
ItemsInCart	0
CouponCode	309
ReferralSource	0
TotalPrice	0